

Bacterial L-Carnitine Catabolism Through 3-Dehydrocarnitine

A review-style synthesis for a molecular-biology audience

1. Executive summary

Many aerobic bacteria — best characterized in the pseudomonads and related α/γ -proteobacteria — can grow on L-carnitine as a sole carbon and/or nitrogen source by dismantling its four-carbon backbone rather than by simply importing carnitine as a compatible solute. The route reviewed here is a compact four-step **assimilatory module**: (1) L-carnitine is imported by a modular ABC transporter; (2) it is oxidized by an NAD^+ -dependent **L-carnitine dehydrogenase (EC 1.1.1.108)** to the β -keto acid **3-dehydrocarnitine**; (3) 3-dehydrocarnitine is cleaved by a **CoA-dependent 3-ketoacid CoA-transferase/thiolase system** that removes a two-carbon (acetyl) unit through a **betainyl-CoA** thioester; and (4) the betainyl-CoA is resolved to **glycine betaine**, which then enters the independent glycine-betaine $\rightarrow$ glycine catabolic pathway.

The overall logic is that of a β -keto-acid chain-shortening reaction grafted onto quaternary-ammonium chemistry: oxidation of the C3 hydroxyl converts an inert β -hydroxy acid into a labile β -keto acid that can be thiolytically shortened by two carbons, converting the 4-carbon carnitine backbone ($\text{N}-\text{C}-\text{C}-\text{C}-\text{COO}^-$) into the 2-carbon betaine backbone ($\text{N}-\text{C}-\text{COO}^-$). The committed oxidation is thermodynamically unfavorable and is pulled forward by the essentially irreversible downstream cleavage. This oxidative, aerobic route is one of three mechanistically distinct ways bacteria attack carnitine, and it must be kept separate from (i) the oxygenolytic **CntAB** route that cleaves carnitine to trimethylamine (TMA), and (ii) the anaerobic **Cai** route that reduces carnitine via crotonobetaine to γ -butyrobetaine. The strongest evidence is genetic (loss-of-function screens in *Pseudomonas aeruginosa*) and enzymological (purified dehydrogenases); the CoA-transferase/thiolase cleavage chemistry and the betainyl-CoA intermediate are inferred from gene annotation, reaction stoichiometry, and precedent, and remain the least directly proven link.

2. Definition and biological boundaries

What is inside the system. The reviewed module is the set of reactions that convert *extracellular L-carnitine* into *intracellular glycine betaine* via *3-dehydrocarnitine*, comprising exactly four functional steps:

1. **Uptake** of L-carnitine across the cytoplasmic membrane.
2. **Oxidation** of L-carnitine to 3-dehydrocarnitine (L-carnitine dehydrogenase, EC 1.1.1.108).
3. **Cleavage** of 3-dehydrocarnitine, with CoA activation of the β -keto acid and thiolytic loss of a C2 unit, generating a betainyl-CoA intermediate.
4. **Glycine-betaine formation** from betainyl-CoA (thioester resolution).

What is outside the system (and frequently confused with it).

- **The oxygenolytic CntAB route $\rightarrow$ TMA.** A two-component Rieske-type oxygenase/reductase (CntA/CntB) cleaves the C–N bond of carnitine to yield trimethylamine plus malic semialdehyde

([P 24591617](#));

biophysics in

([P 32694223](#)).

This is medically prominent (TMA $\rightarrow$ hepatic TMAO $\rightarrow$ cardiovascular risk) but is chemically and enzymologically unrelated to the dehydrogenase route: it destroys the quaternary amine instead of preserving it as betaine.

- **The anaerobic Cai/crotonobetaine route.** In *Escherichia coli* and other Enterobacteriaceae, the *caiTABCDE* operon (anaerobic, CRP/H-NS regulated) converts carnitine via crotonobetaine to γ -butyrobetaine using a carnitine/crotonobetaine dehydratase (CaiB) and a CoA-transferase/ligase (CaiC), with CaiT as the transporter

([P 7815937](#));

([P 8188598](#)).

Confusingly, this route *also* uses CoA thioesters and *also* has a protein called "CaiB," but it is a reductive fermentative pathway that never passes through 3-dehydrocarnitine or glycine betaine.

- **Compatible-solute retention.** Carnitine and glycine betaine are also imported and *retained unmodified* as osmoprotectants (e.g., via BCCT/OpuC transporters;

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Osmoprotective accumulation and catabolic assimilation share transporters but are opposite fates; the reviewed system is the catabolic fate.

- **Downstream glycine-betaine demethylation.** Once glycine betaine is formed, its stepwise demethylation to dimethylglycine → sarcosine → glycine (gbcAB, dgcAB, soxBDAG in *P. aeruginosa*;

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is a separate, independently regulated pathway. It is the *sink* for this module's product but is not part of the module itself.

Competing definitions. The older enzymological literature (reviewed by Kleber,

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frames "bacterial carnitine metabolism" as three routes and treats the dehydrogenase route as *the* assimilatory pathway. Some sources loosely equate "carnitine catabolism" with the gut-microbiome TMA route because of its clinical salience. This review adopts the narrow, mechanistic definition: the oxidative 3-dehydrocarnitine → betaine module.

3. Mechanistic overview

Step 1 — Uptake (obligatory, but not pathway-specific). L-carnitine crosses the membrane before any chemistry occurs. In pseudomonads this is accomplished by the **Cbc ABC transporter**, an unusual system in which a single core permease/ATPase module (**CbcWV**) recruits several interchangeable periplasmic substrate-binding proteins (SBPs). The carnitine-specific SBP **CaiX** binds L-carnitine ($K_m \approx 24 \mu\text{M}$), while choline- and betaine-specific SBPs (CbcX, BetX) serve the same core

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Lower-affinity, broad-specificity BCCT/OpuC transporters can also admit carnitine

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Uptake is therefore obligatory for the pathway but is shared machinery, not a dedicated carnitine importer.

Step 2 — Oxidation (committed step). Cytoplasmic **L-carnitine:NAD⁺ oxidoreductase (EC 1.1.1.108)** oxidizes the C3 hydroxyl of L-carnitine to a ketone, producing 3-dehydrocarnitine (3-oxo-4-trimethylaminobutyrate) and NADH. Purified enzymes from *Agrobacterium* sp. (114 kDa homodimer;

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and *Pseudomonas putida* (62 kDa homodimer;

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are strictly specific for L-carnitine and NAD⁺. The equilibrium lies far toward carnitine ($K_{eq} \approx 2 \times 10^{-12}$ for the analogous D-enzyme;

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so the step is thermodynamically uphill and proceeds only because 3-dehydrocarnitine is rapidly consumed and NADH is reoxidized by respiration. This is the *committed* step of the module.

Step 3 — Cleavage (the defining, rate-committing chemistry). 3-Dehydrocarnitine is a **β -keto acid**: carboxylate at C1, ketone at C3. Two features follow. First, like all β -keto acids it is intrinsically prone to decarboxylation, so it is a reactive, transient intermediate. Second, its C2–C3 bond is activated toward thiolytic (retro-Claisen) cleavage once the C1 carboxylate is converted to a CoA thioester. In *P. aeruginosa*, genetic screening identified **PA1999–PA2000**, the α/β subunits of a predicted **3-ketoacid CoA-transferase**, as required for growth on carnitine and proposed to catalyze "the first step of deacetylation of 3-dhc"

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The most parsimonious chemistry is: CoA transfer onto 3-dehydrocarnitine $\rightarrow$ 3-dehydrocarnitiny-CoA (a β -ketoacyl-CoA) $\rightarrow$ thiolytic cleavage releasing **acetyl-CoA** (from C1–C2) and leaving **betainyl-CoA** (from C3–C4; i.e., trimethylammonio-acetyl-CoA). This removes exactly two carbons and accounts for the carnitine $\rightarrow$ betaine backbone shortening.

Step 4 — Glycine-betaine formation. The betainyl-CoA thioester is resolved to free **glycine betaine** — either by hydrolysis (thioesterase) or, more economically, by CoA transfer back onto an incoming 3-dehydrocarnitine molecule, which would make the CoA-transferase catalytic and couple steps 3 and 4. Glycine betaine is the obligate product: an intact glycine-betaine catabolic pathway is independently required for growth on carnitine

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Obligatory vs. conditional vs. accessory. - *Obligatory*: oxidation (step 2) and cleavage (step 3) — no growth without them. - *Conditional/shared*: uptake (step 1), which can be served by several transporters; the specific importer used depends on the organism and osmotic context. - *Accessory / downstream sink*: betaine demethylation, NADH reoxidation, and acetyl-CoA disposal are required for net assimilation but belong to central metabolism, not the module proper.

4. Major molecular players and active assemblies

Step	Function	Best-characterized players	Notes / evidence
1. Uptake	Import L-carnitine	Cbc ABC transporter : core CbcWV + carnitine-specific SBP CaiX ($K_m \approx 24 \mu\text{M}$); also BCCT/OpuC	Modular SBP recruitment; (P 19919675) (P 17660277)
2. Oxidation	L-carnitine $\rightarrow$ 3-dehydrocarnitine + NADH	L-carnitine dehydrogenase, EC 1.1.1.108 (homodimer; NAD ⁺ -specific)	Purified from <i>Agrobacterium</i> (P 8645721), <i>P. putida</i> (P 3058208); <i>P. aeruginosa</i> homologue in PA5388–PA5384 locus (P 19406895)
3. Cleavage	3-dehydrocarnitine (+CoA) $\rightarrow$ betainyl-CoA + acetyl-CoA	3-ketoacid CoA-transferase (α/β subunits, <i>P. aeruginosa</i> PA1999–PA2000) + thiolase activity	Genetically required; enzymology inferred (P 19406895)
4. Betaine formation	betainyl-CoA $\rightarrow$ glycine betaine	Thioesterase or CoA-transfer resolution (enzyme not definitively assigned)	Glycine betaine confirmed as obligate product (P 19406895)
Regulation	Substrate-responsive induction	<i>P. aeruginosa</i> PA5389 (induces dehydrogenase locus in response to carnitine) and PA1998 (induces PA1999–PA2000)	Transcription-factor pair; (P 19406895)
Product sink (outside module)	Glycine betaine $\rightarrow$ glycine	GbcAB, DgcAB, SoxBDAG; master regulator GbdR	(P 17951379) (P 24097953)

Regulatory architecture. The two catabolic loci in *P. aeruginosa* are independently induced: PA5389 drives the dehydrogenase region in response to carnitine, and PA1998 drives the CoA-transferase region

([P 19406895](#)).

Downstream, the AraC-family regulator **GbdR** integrates the betaine node, inducing betaine/DMG/sarcosine catabolism *and* the quaternary-amine transporters (BetX, CbcXWV) in response to glycine betaine and dimethylglycine

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with SouR providing sarcosine-specific control

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Thus carnitine assimilation is wired as a relay of substrate-responsive AraC/other regulators rather than a single operon.

5. Evolutionary and cell-biological variation

Lineage distribution. The oxidative dehydrogenase route is documented across diverse aerobic proteobacteria — *Pseudomonas* (putida, aeruginosa), *Agrobacterium/Rhizobium*, and other soil/plant-associated organisms

([P 9037756](#)).

The Cbc/CaiX uptake system is shared between *P. aeruginosa* and the plant pathogen *P. syringae*

([P 19919675](#)),

indicating conservation of at least the uptake and probably the catabolic logic across the genus.

Alternative routes to overlapping outcomes. Different lineages have evolved biochemically unrelated solutions to the same substrate: - *Oxidative assimilation* (this module) — pseudomonads/rhizobia, preserves the quaternary amine and funnels it to glycine. - *Oxygenolytic C–N cleavage to TMA* — *Acinetobacter*, gut microbiota; CntAB Rieske oxygenase

([P 24591617](#)).

Destroys the trimethylammonium head group. - *Anaerobic reduction to γ -butyrobetaine* — Enterobacteriaceae; Cai system

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A redox-sink fermentative use. These are best viewed as convergent, ecologically partitioned strategies (aerobic soil catabolism vs. anaerobic gut vs. facultative), not variants of one pathway.

Physiological-state variation. Because carnitine and glycine betaine are also potent osmoprotectants, the same imported molecule can be *catabolized* or *retained* depending on osmotic stress, nitrogen/carbon limitation, and regulator state. Under high osmolarity, retention (BCCT/OpuC accumulation) is favored; under nutrient limitation with low osmotic stress, catabolic assimilation dominates. This is a genuine state-dependent fork rather than a fixed fate.

Host relevance. In *P. aeruginosa*, host-derived quaternary amines (choline, carnitine, glycine betaine) are abundant nutrients during infection, and their catabolism is linked (via GbdR) to expression of virulence-associated phospholipase C

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tying this catabolic module to the organism's nutritional adaptation in host tissue.

6. Constraints, dependencies, and failure modes

Mandatory ordering. The steps are strictly sequential and each depends on the prior product: uptake → oxidation → cleavage → betaine formation. Oxidation *must* precede cleavage because the cleavage chemistry requires the C3 **ketone**; the parent β-hydroxy acid (carnitine) cannot undergo thiolytic C2 shortening. This is the key chemical constraint that rules out a direct carnitine → betaine conversion without prior oxidation.

Thermodynamic pull. The committed oxidation is strongly endergonic ($K_{eq} \approx 10^{-12}$;

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The pathway can run only if (i) 3-dehydrocarnitine is consumed rapidly by an essentially

irreversible cleavage and (ii) NADH is reoxidized — i.e., the route is effectively **aerobic/respiratory**. This explains why the assimilatory route is found in aerobes and why the alternative anaerobic fate of carnitine uses an entirely different (reductive) chemistry.

Intermediate instability. 3-Dehydrocarnitine is a β -keto acid prone to spontaneous decarboxylation; efficient channeling to the CoA-transferase is likely needed to avoid loss of the intermediate. This is a plausible driver for coupling the dehydrogenase and CoA-transferase steps and for co-regulation of the two loci.

Compartment/co-substrate constraints. All chemistry after uptake is cytoplasmic and requires NAD^+ and CoA/acyl-CoA pools; the acetyl-CoA released feeds central metabolism, and the CoA must be recycled (via transfer or thioesterase) for the pathway to be catalytic. Failure to reoxidize NADH or to recycle CoA would stall the module.

Failure modes / genetic evidence. Loss of the dehydrogenase locus (PA5388–PA5384), the CoA-transferase locus (PA1999–PA2000), their regulators (PA5389, PA1998), or the downstream glycine-betaine catabolic genes each abolishes growth on carnitine in *P. aeruginosa* (

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confirming that all four functional steps plus the betaine sink are jointly required for assimilation.

7. Controversies and open questions

1. **Direct enzymology of the cleavage step.** The 3-ketoacid CoA-transferase assignment (PA1999–PA2000) rests on gene annotation, mutant phenotypes, and reaction logic (

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the enzyme has not, to the standard of the purified dehydrogenases, been shown *in vitro* to convert 3-dehydrocarnitine to betainyl-CoA + acetyl-CoA. Whether a separate thiolase completes the thiolysis, and whether the "CoA-transferase" is truly a transferase (recycling CoA) or a ligase, remains to be nailed down.

2. **Existence and isolation of betainyl-CoA.** The betainyl-CoA intermediate is chemically reasonable and is named in pathway schemes, but its direct isolation and characterization from this route are, to our reading, not firmly established. This is the single most important experimental gap.

3. **CoA donor/acceptor identity.** If step 3 is a CoA-transferase, what is the physiological CoA donor (succinyl-CoA? acetoacetyl-CoA? the betainyl-CoA of the previous cycle)? Coupling steps 3 and 4 as a single catalytic CoA relay is attractive but unproven.
4. **How the enzyme escapes decarboxylation of the β -keto acid.** Mechanistically, channeling or a specific binding mode must suppress non-productive decarboxylation of 3-dehydrocarnitine; this has not been structurally addressed.
5. **Generality beyond pseudomonads.** Much of the *pathway-level genetic* evidence is from *P. aeruginosa*, while much of the *enzyme* evidence is from *Agrobacterium* and *P. putida*. Whether the same gene complement and regulatory wiring operate across all organisms said to use "the dehydrogenase route" is inferred, not comprehensively demonstrated — a caution against overgeneralizing from one organism.
6. **Evolutionary origin.** The dehydrogenase belongs to the broad short-chain-dehydrogenase-like landscape of NAD⁺ oxidoreductases, and the cleavage enzyme to the ancient, widely distributed **CoA-transferase/thiolase** superfamilies used throughout β -keto-acid and fatty-acid metabolism. The most plausible reading is that this module is a **repurposing of pre-existing β -keto-acid chain-shortening chemistry** to quaternary-ammonium substrates, rather than a de novo invention. Identifying the closest non-carnitine CoA-transferase/thiolase relatives (e.g., generic 3-oxoacid CoA-transferases, SCOT/OXCT-type enzymes) would be the best way to reconstruct the ancestral role; this phylogenetic analysis has not been done specifically for the carnitine enzymes.

8. Key references

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Evidence-confidence key

- **Strong (purified enzyme / genetic loss-of-function):** committed oxidation by L-carnitine dehydrogenase; requirement of the dehydrogenase and CoA-transferase loci and of the betaine sink for growth; identity/affinity of the Cbc/CaiX uptake system.
 - **Moderate (annotation + phenotype + reaction logic):** the CoA-transferase as the cleavage catalyst; two-carbon (acetyl-CoA) loss; glycine betaine as product.
 - **Inferred / open:** the discrete betainyl-CoA intermediate and its isolation; the CoA donor/acceptor and step-3/step-4 coupling; cross-lineage generality of the exact gene complement; the pathway's detailed phylogenetic origin.
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